

# Lecture 05: Hydrogen atom

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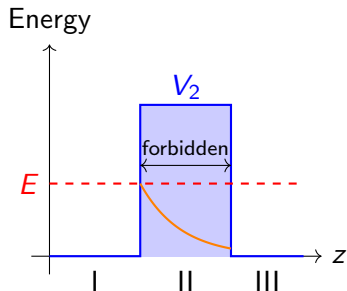
MAT SCI 120 — Winter 2026

## Where we stopped

- ▶ We have seen the interpretation of wave-particle duality of electrons through the lens of wavefunctions  $\psi$
- ▶ The Schrödinger's equation describes the dependence of  $\psi$  with space and time.
- ▶ We started using this equation to solve a few examples: free electron, potentials, etc.

## Electrons and potential barriers: when $E < V$

- ▶ Classical particle cannot exist where kinetic energy would be negative ( $E < V$ )
- ▶ Quantum mechanics: wave function doesn't vanish—it **decays exponentially**
- ▶ Region where  $E < V_2$  is called the **classically forbidden region**
- ▶ Probability density  $|\psi|^2$  remains finite inside barrier



## Wave function inside the barrier

**Schrödinger equation inside barrier:**

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dz^2} + V_2\psi = E\psi \quad (1)$$

**Rearranging:**

$$\frac{d^2\psi}{dz^2} = \frac{2m}{\hbar^2}(V_2 - E)\psi = \gamma^2\psi \quad \text{where } \gamma^2 > 0 \quad (2)$$

**Decay Constant in Barrier**

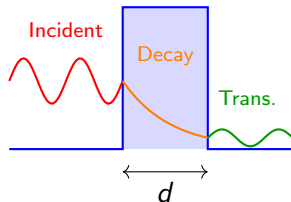
$$\gamma = \sqrt{\frac{2m(V_2 - E)}{\hbar^2}} \quad (3)$$

**Solution:** Real exponentials (not oscillatory)

$$\psi_2 = Ae^{-\gamma z} + Be^{+\gamma z} \quad (4)$$

# Exponential decay through finite barrier

- ▶ For barrier of width  $d$ : keep both exponential terms inside
- ▶ Wave function decays as  $e^{-\gamma z}$  through barrier
- ▶ Amplitude at exit reduced by factor  $e^{-\gamma d}$
- ▶ Probability reduced by  $e^{-2\gamma d}$

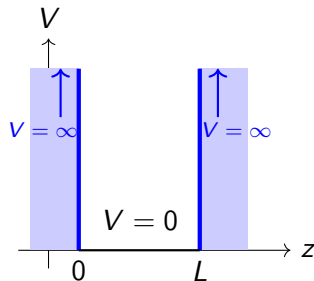


## Amplitude Decay

$$\psi_2(z) \propto e^{-\gamma z}, \quad |\psi|^2 \propto e^{-2\gamma d}$$

## Infinite potential well: problem setup

- ▶ Electron confined between rigid walls at  $z = 0$  and  $z = L$  with  $V = \infty$  outside
- ▶ Inside well ( $0 < z < L$ ):  $V = 0$ , electron moves freely
- ▶ **Boundary conditions:**  $\psi$  must vanish at walls
- ▶ Simplest exactly solvable quantum confinement problem



## Solving inside the infinite well

**Time-independent Schrödinger equation with  $V = 0$ :**

Wave Equation Inside Well

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dz^2} = E\psi \quad (5)$$

**Rearranging:**

$$\frac{d^2\psi}{dz^2} = -\frac{2mE}{\hbar^2}\psi = -k^2\psi \quad (6)$$

**General solution:**

$$\psi(z) = A \sin(kz) + B \cos(kz) \quad (7)$$

**Wave vector related to energy:**

$$k^2 = \frac{2mE}{\hbar^2} \Rightarrow E = \frac{\hbar^2 k^2}{2m} \quad (8)$$

## Applying boundary conditions

**At  $z = 0$ :**  $\psi(0) = 0$

$$A \sin(0) + B \cos(0) = B = 0 \quad \Rightarrow \quad \psi(z) = A \sin(kz) \quad (9)$$

**At  $z = L$ :**  $\psi(L) = 0$

$$A \sin(kL) = 0 \quad \Rightarrow \quad kL = n\pi \quad (10)$$

### Quantization Condition

$$k_n = \frac{n\pi}{L}, \quad n = 1, 2, 3, \dots \quad (11)$$

- ▶ **Quantization emerges:** only discrete  $k$  values allowed
- ▶ Quantum number  $n = 1, 2, 3, \dots$  (note:  $n = 0$  gives trivial  $\psi = 0$ )



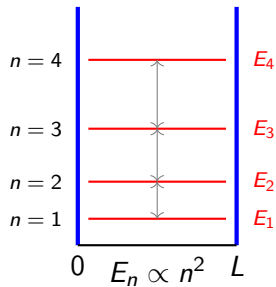
# Quantized energy levels in infinite well

Substituting  $k = n\pi/L$  into  $E = \hbar^2 k^2 / 2m$ :

## Energy Eigenvalues

$$E_n = \frac{h^2 n^2}{8mL^2} = \frac{\hbar^2 \pi^2 n^2}{2mL^2}$$

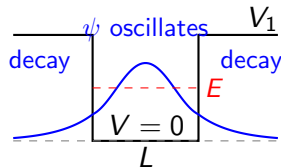
- ▶ Ground state ( $n = 1$ ) has non-zero energy:  
**zero-point energy**
- ▶ Energy spacing increases with  $n$
- ▶ Narrower wells  $\Rightarrow$  higher energies



# Finite potential well: realistic confinement

## Key Features:

- ▶ Finite barrier height  $V_1$  allows wave function penetration into barriers
- ▶ Inside well: oscillatory solution with  $k^2 = \frac{2mE}{\hbar^2}$
- ▶ Outside well: exponential decay with  $\gamma^2 = \frac{2m(V_1 - E)}{\hbar^2}$
- ▶ Bound states exist only for  $E < V_1$ ; **finite number** of discrete levels



## Transcendental equation and graphical solution

**Matching boundary conditions** ( $\psi$  and  $d\psi/dz$  continuous) yields:

$$k \tan\left(\frac{kL}{2}\right) = \gamma \quad (12)$$

where  $k = \sqrt{2mE}/\hbar$  and  $\gamma = \sqrt{2m(V_1 - E)}/\hbar$ .

- ▶ No closed-form solution
- ▶ Graphical: plot both sides vs  $E$
- ▶ Intersections  $\rightarrow$  allowed energies
- ▶ Deeper/wider wells  $\rightarrow$  more states
- ▶ You do not need to solve this in a midterm — but you will see this in the homework!

## Pause

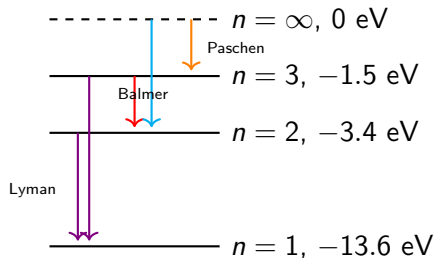
We have learned how to use the Schrödinger's equation to solve toy examples.  
Can we use this equation to solve a “real-world” example?

# Remembering the Bohr model

- ▶ Postulated quantized angular momentum:  $L = n\hbar$  for integer  $n$
- ▶ Correctly predicted hydrogen energy levels
- ▶ Explained Rydberg formula for spectral lines and ionization energy

## Bohr energy levels:

$$E_n = -\frac{13.6 \text{ eV}}{n^2}, \quad n = 1, 2, 3, \dots$$



## Bohr model: critical failures

- ▶ Cannot explain multi-electron atoms (He, Li, ...) — no systematic extension
- ▶ Fails to predict spectral line intensities and fine structure splitting
- ▶ Violates uncertainty principle: electron cannot have definite orbit and momentum simultaneously

### Heisenberg uncertainty principle:

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

### Key Insight

Sharp circular orbit implies  $\Delta x \rightarrow 0$  along radial direction, requiring  $\Delta p \rightarrow \infty$  — unphysical!

# Setting up the hydrogen atom problem

- ▶ Single electron bound to proton by Coulomb attraction
- ▶ Spherically symmetric potential  $\rightarrow$  wave function depends only on radius  $r$
- ▶ Goal: Find allowed energies  $E$  and wave functions  $\psi(r)$

## Coulomb potential:

$$V(r) = -\frac{e^2}{4\pi\epsilon_0 r}$$

*Notation:*  $e$  = elementary charge,  $\epsilon_0$  = permittivity of free space

# Time-independent Schrödinger equation for hydrogen

- ▶ For spherically symmetric  $\psi(r)$ , Laplacian simplifies to radial derivatives only
- ▶ Substitute Coulomb potential into Schrödinger equation
- ▶ Seek bound state solutions with  $E < 0$  (electron confined to atom)

## Radial Schrödinger equation:

$$-\frac{\hbar^2}{2m} \left( \frac{d^2\psi}{dr^2} + \frac{2}{r} \frac{d\psi}{dr} \right) - \frac{e^2}{4\pi\epsilon_0 r} \psi = E\psi$$

*Notation:*  $m$  = electron mass,  $\hbar$  = reduced Planck constant,  $\psi$  = wave function



## Trial solution: exponential wave function

- ▶ Physical intuition:  $\psi$  must decay at large  $r$  (bound state) and be finite at  $r = 0$
- ▶ Try simplest decaying form with unknown constant  $c_0$
- ▶ Compute derivatives for substitution

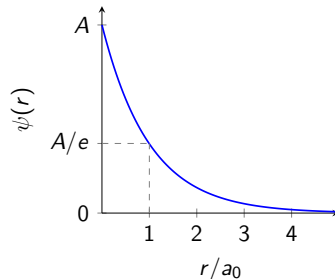
### Trial wave function:

$$\psi(r) = Ae^{-c_0 r}$$

Derivatives:

$$\frac{d\psi}{dr} = -c_0\psi$$

$$\frac{d^2\psi}{dr^2} = c_0^2\psi$$



## Matching coefficients to find $c_0$ and $E$

Substitute  $\psi = Ae^{-c_0 r}$  into Schrödinger equation:

$$-\frac{\hbar^2}{2m} \left( c_0^2 \psi - \frac{2c_0}{r} \psi \right) - \frac{e^2}{4\pi\epsilon_0 r} \psi = E\psi$$

Rearranging:

$$\underbrace{-\frac{\hbar^2 c_0^2}{2m} \psi}_{\text{constant}} + \underbrace{\frac{1}{r} \left( \frac{\hbar^2 c_0}{m} - \frac{e^2}{4\pi\epsilon_0} \right) \psi}_{\text{must vanish}} = E\psi$$

**Matching coefficients:**

- ▶ Coefficient of  $1/r$ :  $c_0 = \frac{me^2}{4\pi\epsilon_0 \hbar^2} = \frac{1}{a_0}$
- ▶ Constant term:  $E_1 = -\frac{\hbar^2 c_0^2}{2m} = -13.6 \text{ eV}$

**Bohr radius:**

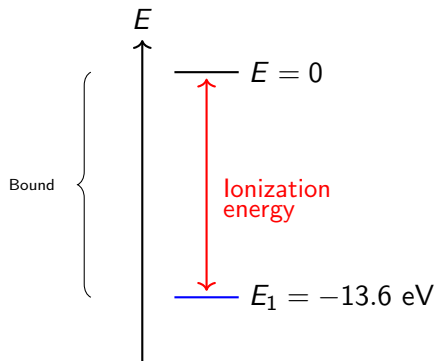
$$a_0 = \frac{4\pi\epsilon_0 \hbar^2}{me^2} = 0.0528 \text{ nm}$$

## Ground state energy: the $-13.6$ eV result

- ▶ Ground state energy matches experimental ionization energy **exactly**
- ▶ Negative energy confirms bound state;  $|E_1|$  is ionization energy
- ▶ Quantum mechanics reproduces Bohr result with proper wave interpretation

### Ground state energy:

$$E_1 = -\frac{me^4}{8\epsilon_0^2 h^2} = -13.6 \text{ eV}$$



## Hydrogen energy depends only on $n$

### Key Result from Schrödinger Equation:

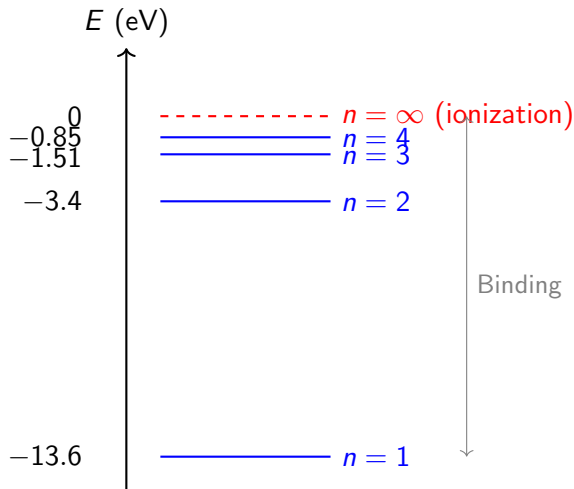
$$E_n = -13.6 \frac{1}{n^2} \text{ eV}, \quad n = 1, 2, 3, \dots \quad (13)$$

### Physical Interpretation:

- ▶ Energy is **negative**  $\Rightarrow$  bound state (electron trapped by nucleus)
- ▶ Energy scales as  $1/n^2 \Rightarrow$  levels converge toward ionization
- ▶ **Degeneracy:** All states with same  $n$  have identical energy

**Notation:**  $E_n$  = energy of level  $n$ ; negative sign = bound below continuum

## Energy level diagram for hydrogen

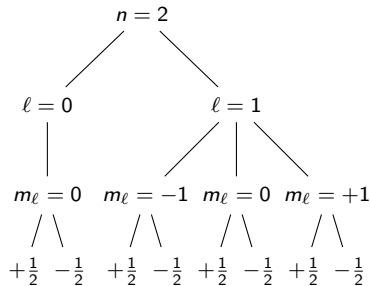


- ▶ Levels converge toward  $E = 0$  (continuum)
- ▶ **Degeneracy:**  $n^2$  orbital states share same energy (ignoring spin)

# In general: quantum numbers

## Quantum Numbers:

- ▶ Principal  $n = 1, 2, 3, \dots$   
determines energy shell
- ▶ Orbital angular momentum  
 $\ell = 0, 1, \dots, n-1$   
determines orbital shape
- ▶ Magnetic  $m_\ell = -\ell, \dots, 0, \dots, +\ell$   
determines spatial orientation
- ▶ Spin  $s = \pm \frac{1}{2}$   
determines intrinsic angular momentum



Example:  $n = 2$  hierarchy

# Spectroscopic Notation: s, p, d, f Orbitals

## Historical Letter Designations:

- ▶  $l = 0 \rightarrow$  **s** orbital (sharp)
- ▶  $l = 1 \rightarrow$  **p** orbital (principal)
- ▶  $l = 2 \rightarrow$  **d** orbital (diffuse)
- ▶  $l = 3 \rightarrow$  **f** orbital (fundamental)

| $l$ | Letter | Example    |
|-----|--------|------------|
| 0   | s      | 1s, 2s, 3s |
| 1   | p      | 2p, 3p, 4p |
| 2   | d      | 3d, 4d, 5d |
| 3   | f      | 4f, 5f     |

**Full notation:**  $n + \text{letter}$

**Note:** Names originated from spectral line characteristics in atomic emission spectra

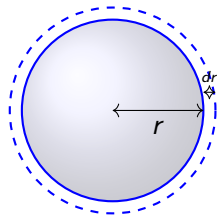
# Why radial probability density uses $r^2|\psi|^2$

## 3D Probability in Spherical Shell:

$$P(r) dr = |\psi|^2 \times 4\pi r^2 dr \quad (14)$$

- ▶ Volume element grows as  $r^2$
- ▶ Radial probability density:  $r^2|\psi|^2$
- ▶ Maximum of  $r^2|\psi|^2$  gives **most probable radius**
- ▶ This differs from maximum of  $|\psi|^2$ !

**Key:** The  $r^2$  weighting captures where electrons actually reside in 3D space



$$\text{Volume} = 4\pi r^2 dr$$



## Ground state 1s: most probable radius = Bohr radius

**For 1s state:**  $\psi_{1s} \propto e^{-r/a_0}$

Radial probability density:  $r^2|\psi|^2 \propto r^2 e^{-2r/a_0}$

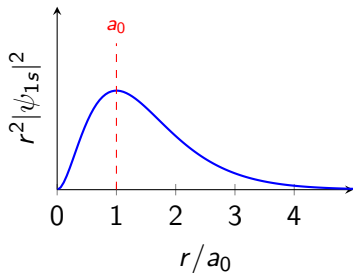
**Finding the maximum:**

$$\frac{d}{dr} \left( r^2 e^{-2r/a_0} \right) = 0$$

$$r_{\text{mp}} = a_0 = 0.0528 \text{ nm}$$

(15)

- ▶ QM recovers Bohr radius!
- ▶ Mean radius:  $\langle r \rangle = 1.5 a_0$



## Radial nodes: where probability vanishes

**Definition:** Radial nodes are spherical surfaces where  $\psi(r) = 0$

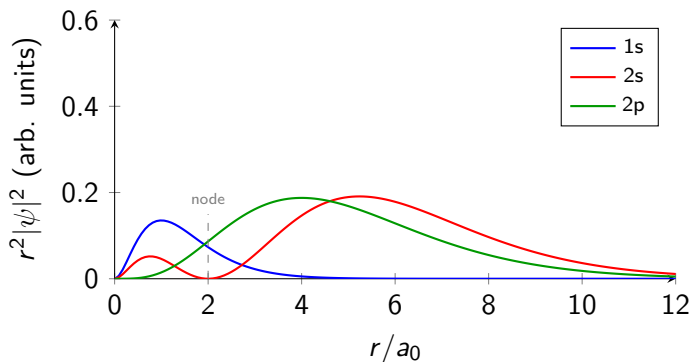
$$N_{\text{nodes}} = n - \ell - 1 \quad (16)$$

### Examples:

- ▶ **1s** ( $n = 1, \ell = 0$ ):  $1 - 0 - 1 = 0$  nodes
- ▶ **2s** ( $n = 2, \ell = 0$ ):  $2 - 0 - 1 = 1$  node
- ▶ **2p** ( $n = 2, \ell = 1$ ):  $2 - 1 - 1 = 0$  radial nodes (but has angular node)
- ▶ **3s** ( $n = 3, \ell = 0$ ):  $3 - 0 - 1 = 2$  nodes

**Physical meaning:** Nodes separate regions where wave function changes sign

## Comparing 1s, 2s, and 2p Radial Distributions



- ▶ 2s has node at  $r \approx 2a_0$ ; **closer** to nucleus than 2p
- ▶ Higher  $n$  states extend further: **shell structure emerges**

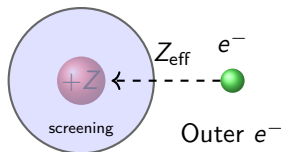
# The Pauli exclusion principle

## Fundamental Rule:

- ▶ No two electrons can occupy the same quantum state  $(n, \ell, m_\ell, s)$
- ▶ Each orbital  $(n, \ell, m_\ell)$  holds at most 2 electrons with opposite spins
- ▶ Fundamental principle governing all multi-electron systems
- ▶ Explains why electrons fill shells rather than collapse to ground state

## Effective nuclear charge: screening by inner electrons

- ▶ Outer electrons don't see full nuclear charge  $Z$  due to inner electron shielding
- ▶ Effective nuclear charge:  $Z_{\text{eff}} = Z - \sigma$ 
  - $\sigma$  = screening constant
- ▶ Inner shells screen more effectively than electrons in same shell
- ▶  $Z_{\text{eff}}$  increases across a period



Inner electrons

## Energy with Effective Nuclear Charge

$$E_n \approx -13.6 \frac{Z_{\text{eff}}^2}{n^2} \text{ eV} \quad (17)$$

## Key concepts recap: quantum numbers to periodicity

### Core principles connecting atomic structure to chemistry:

- ▶ Four quantum numbers ( $n, \ell, m_\ell, s$ ) completely specify electron state
- ▶ Pauli exclusion  $\rightarrow$  shell structure  $\rightarrow$  periodic table organization
- ▶ Electron-electron interactions cause  $\ell$ -dependent energy ordering

### Energy of hydrogen-like atoms

$$E_n = -13.6 \frac{Z_{\text{eff}}^2}{n^2} \text{ eV}$$

- ▶  $Z_{\text{eff}}$  = effective nuclear charge (screened by inner electrons)
- ▶  $n$  = principal quantum number

## Recap: quantum numbers and degeneracy

### Test your understanding:

**Q1:** For  $n = 3$ , what are the allowed values of  $\ell$  and corresponding orbital names?

**Q2:** Calculate  $E_3$  and  $E_4$  using  $E_n = -13.6/n^2$  eV

**Q3:** Why do 3s, 3p, and 3d all have the same energy in hydrogen?

**Q4:** How many distinct  $(n, \ell, m_\ell)$  states exist for  $n = 2$ ?